29 October 2026

Dear Editors-in-Chief,

Please find enclosed the manuscript "A Lightweight Detection Pipeline for Earthquake Early Warning and Its Evaluation on Three Seismic Networks", which we submit for exclusive consideration as a research article in Computers & Geosciences. We confirm that the submission follows the journal's requirements and includes every item of its submission checklist.

The manuscript addresses both of the journal's scopes. Computationally, it presents ayzek, a self-contained pipeline for the detection stage of earthquake early warning that also reports a P-wave displacement magnitude. Its hand-written inference reproduces the PyTorch networks to within floating-point rounding, its replays of archived data are deterministic, and the same build runs on a Raspberry Pi 4. It also presents a replay protocol, implemented as a reproducible workflow, that measures detectors at their operating point with chance correction, paired tests and catalog-quiet data, whose settings, including the baselines', were fixed in a public, timestamped release before any test waveform was fetched. Geoscientifically, it measures how a learned detector, two classical triggers and two published models behave on continuous records of three networks under that protocol: the Greek networks served by the National Observatory of Athens and New Zealand's GeoNet, which the learned detector had not seen, and the Turkish KO network the detector was trained on, held out in time. At the settings matched on unmatched alarms the learned detector detected 22.4 more chance-corrected events than STA/LTA of 177 on the Greek sequences, though with twice its unmatched alarms, and 18.1 more of 151 on the New Zealand sequences, with equal unmatched alarms; the KO test, under its target in its two-month pool, was not significant. Causal PhaseNet was equivalent to the learned detector on both unseen networks, so the gain is that of learned detectors over an amplitude trigger rather than of ours. On the catalog-quiet windows the learned detector's alarm rate was not distinguishable from STA/LTA's on New Zealand or Greece, where an exploratory analysis traces half of its alarms to one station's emergent noise that STA/LTA ignores, and its alarms came 1.2 to 2.9 s later. On a 728-day record at one station, two detectors with nearly equal ROC-AUC differed by a factor of 1.6 in recall at one false alarm per day.

The source code, model weights, the protocol and the test settings, and the reproduction scripts are in a public repository under the GNU General Public License, archived on Zenodo, with the details listed in the section "Code availability". The test waveforms are open at their data centers and are not redistributed; the released request records and file hashes let anyone fetch and check the same data. The AFAD development waveforms cannot be redistributed, and public KOERI datasets reproduce the development results.

The use of generative AI tools during this work is declared in the manuscript, and the authors reviewed and edited all content and take full responsibility for it. The manuscript has not been published and is not under consideration elsewhere, all authors approve its submission, and they declare no competing interests.

Sincerely,

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